

Static and Behavioral Analysis of a PUA-Classified DLL

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Project Description

This report documents an independent malware analysis conducted on RUNE64.dll, a 64-bit dynamic-link library extracted from a pirated copy of Grand Theft Auto V (version 2025.03.19). Windows Defender flagged the binary under multiple threat classifications — Severe through Low — preventing the game from launching. The objective was to determine whether those detections constituted genuine threats or false positives by applying three complementary methodologies: dynamic behavioral sandbox analysis via VirusTotal, Windows Defender quarantine auditing and file restoration using the MpCmdRun.exe command-line utility, and static Portable Executable (PE) structure analysis using Detect It Easy (DIE). The SHA-256 hash of the primary sample:

```
a184efddc60f12f159131c1a49662460ea766b1d0960423df723ad984d966d6b
```

Task 1: Behavioral Sandbox Analysis via VirusTotal

The SHA-256 hash of RUNE64.dll was submitted to VirusTotal for automated sandbox execution. Three environments analyzed the binary — C2AE, CAPE Sandbox, and Zenbox. The resulting Activity Summary reported 6 LOW and 5 INFO-level MITRE ATT&CK signatures, with no IDS rules, Sigma rules, or dropped files detected. The only recorded network activity consisted of 5 DNS resolutions. Behavioral tags assigned to the sample were *idle* and *obfuscated*.

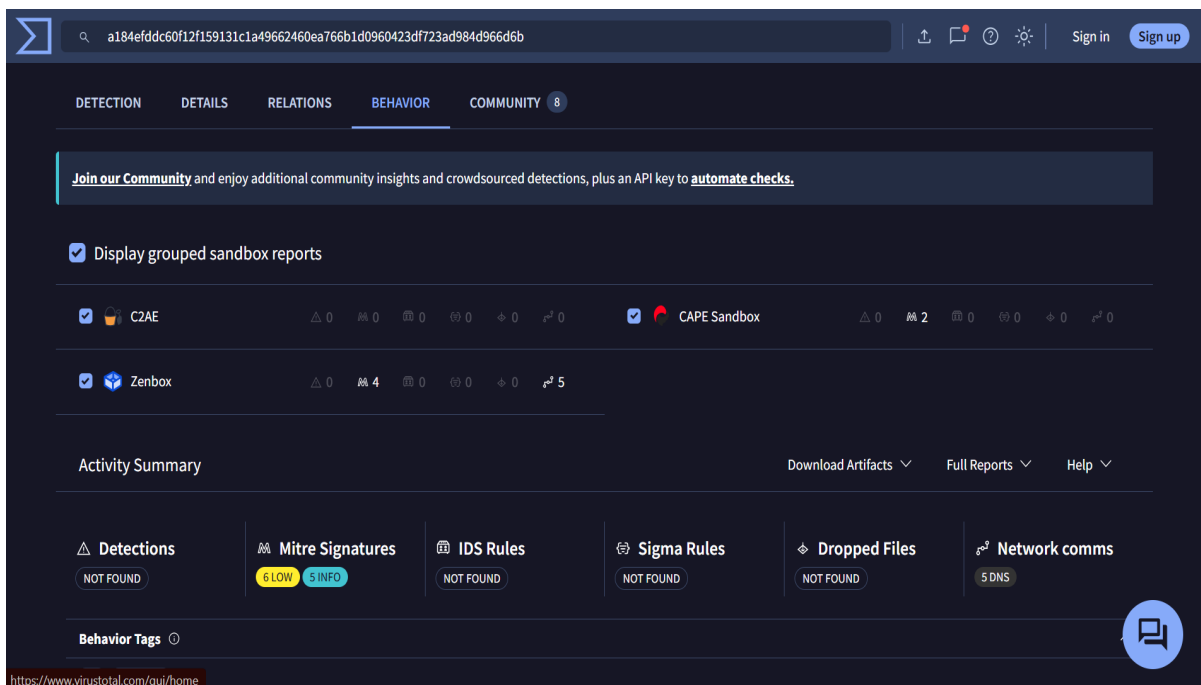


Figure 1: VirusTotal Activity Summary — sandbox engines, MITRE signature severity counts, and DNS communication overview

The MITRE ATT&CK mapping produced detections across four tactical categories. Execution (TA0002) flagged Hijack Execution Flow (T1574), consistent with rundll32.exe loading the DLL via its RUNE named export. Privilege Escalation (TA0004) identified Abuse Elevation Control Mechanism (T1548). The Stealth/Defense Evasion tactic (TA0005) returned the highest technique count: Obfuscated Files or Information (T1027), System Binary Proxy Execution (T1218) via rundll32.exe, and a second Hijack Execution Flow entry. Discovery (TA0007) mapped Process Discovery (T1057) and System Information Discovery (T1082), both attributable to WMI hardware enumeration during game engine initialization.

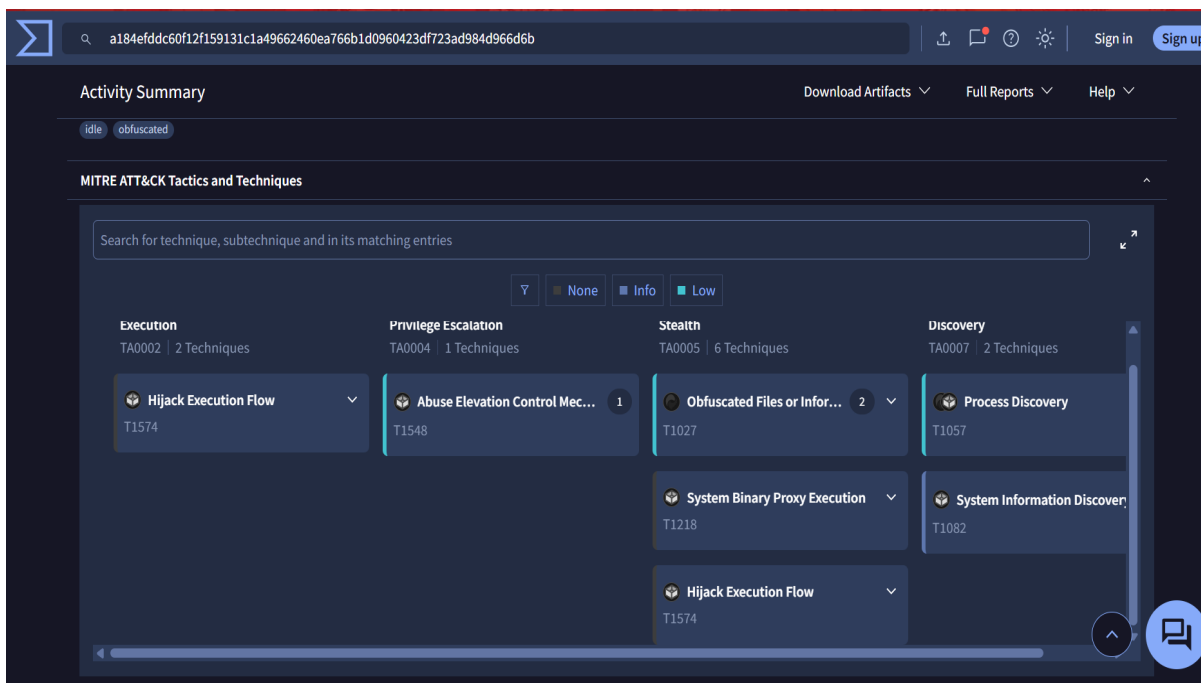


Figure 2: MITRE ATT&CK Tactics and Techniques — Execution, Privilege Escalation, Stealth, and Discovery

The Malware Behavior Catalog classified observed behaviors under Anti-Behavioral Analysis (OB0001), Anti-Static Analysis (OB0002), Defense Evasion (OB0006), Discovery (OB0007), and Communication (OC0006). All 5 DNS resolutions resolved exclusively to Microsoft CDN and telemetry infrastructure: a1666.dscr.akamai.net, assets.msn.com, assets.msn.com-ion.edgesuite.net, ecs-office.s-0005.dual-s-msedge.net, and s-0005.dual-s-msedge.net. This resolution pattern is characteristic of standard Windows OS communication rather than attacker-controlled Command and Control (C2) channels.

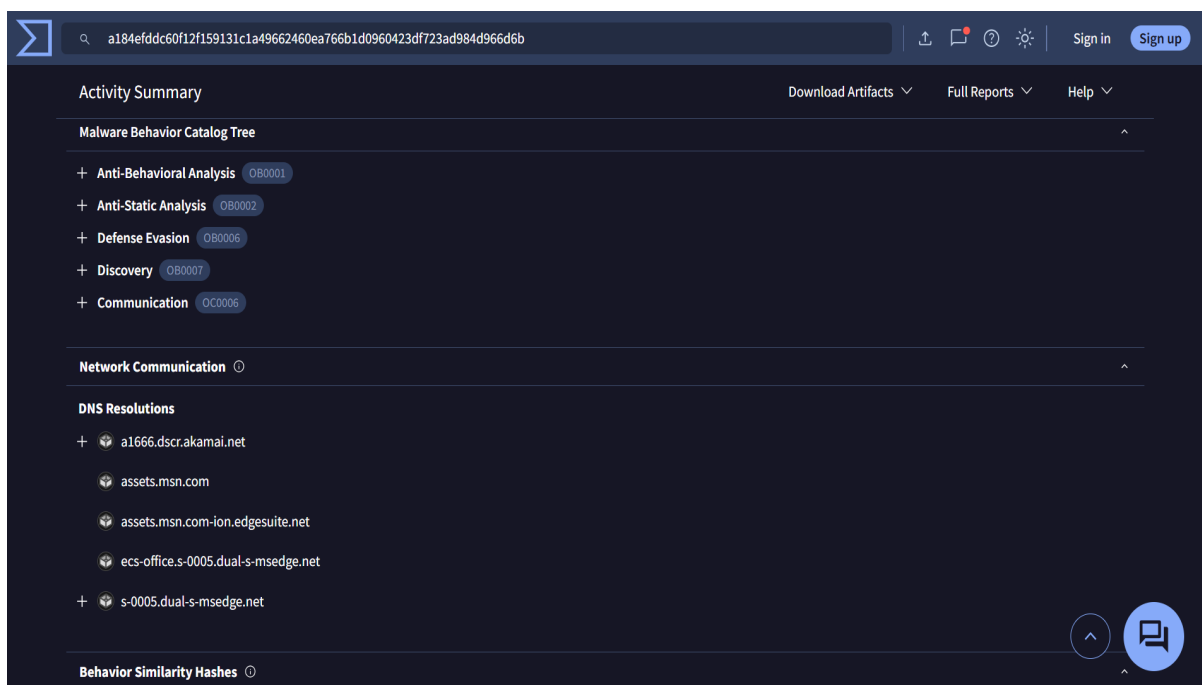


Figure 3: Malware Behavior Catalog classification tree and DNS resolution log

File system interactions showed the binary opening multiple manifest variants of itself across several sandbox-assigned filenames — attachment.dll (the C2AE sandbox's name for the uploaded sample), library.dll, and readme.dll. These are not game runtime dependencies being verified; they are sandbox-generated artifacts produced as the analysis environment brute-forced the DLL's export ordinals (reflected in the shell command %SANDBOX_DLL_LOADER_AMD64% %SAMPLEPATH% %WORKDIR% 483, where 483 denotes the number of export ordinals tested) and re-executed the binary under renamed copies. Oracle Java path queries and manifest lookups are a byproduct of Windows' side-by-side assembly resolution triggered during each rundll32.exe invocation. The only file write operation targeted \Device\ConDrv\Connect, the Windows console device interface, indicating console output rather than disk-level persistence. Registry interactions were confined to standard Windows compatibility layer keys: AppCompatFlags\Layers for application shim evaluation, Explorer\Shell Folders for default path resolution, and AppModel\Lookaside entries for runtime package lookup. No Run keys, scheduled task registrations, or service installation entries were observed.

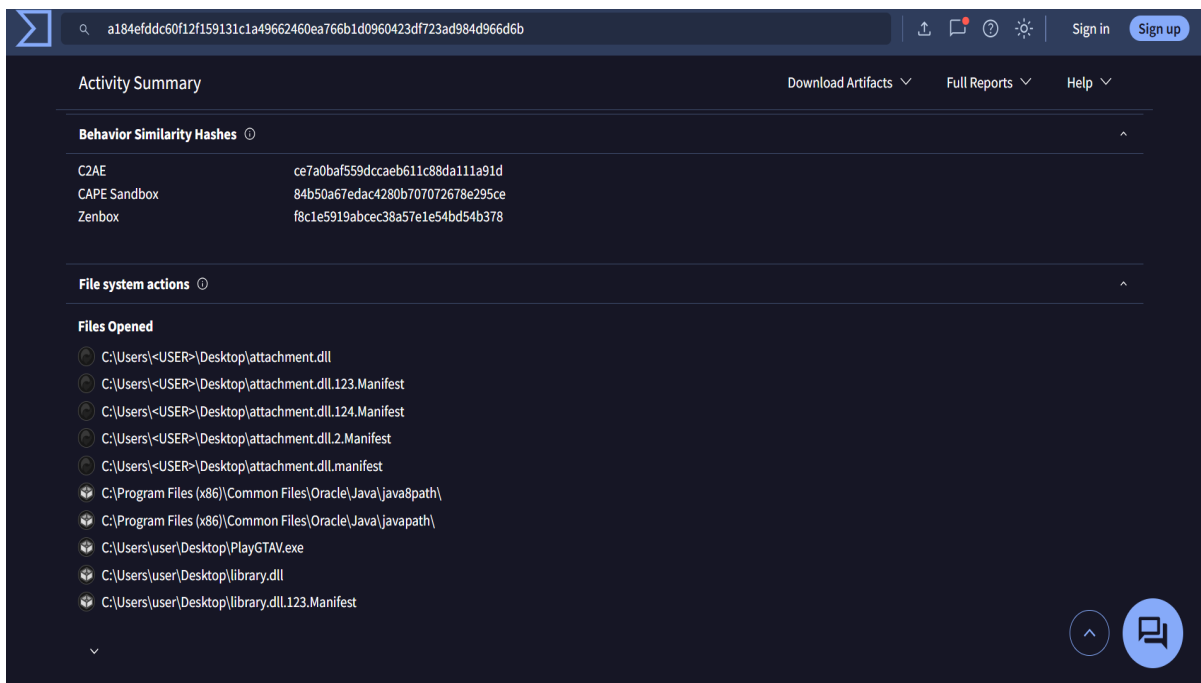


Figure 4: File system actions — sandbox-generated DLL filenames (attachment.dll, library.dll) and manifest variants opened during export ordinal enumeration

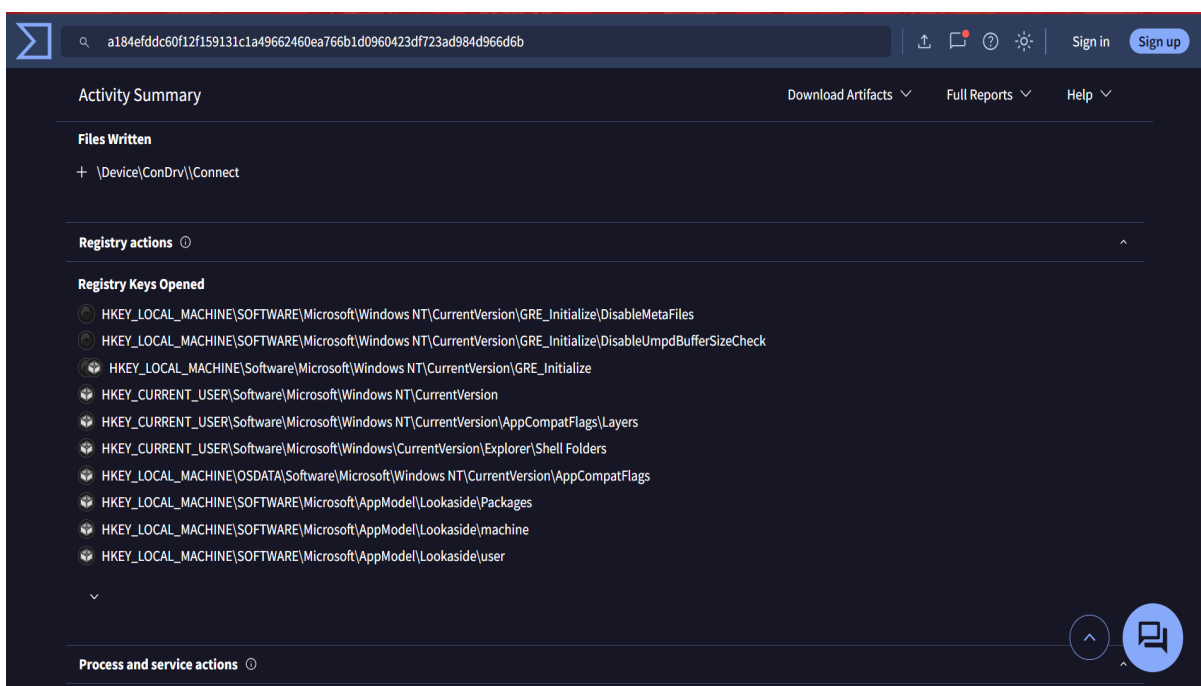


Figure 5: Files written and registry keys opened — \Device\ConDrv\Connect as sole write target and standard Windows compatibility layer keys confirming absence of persistence

The process execution tree confirmed the expected invocation chain for a scene crack. The sandbox loader invoked rundll32.exe with the RUNE export as the entry point. loadll64.exe

subsequently spawned cmd.exe, which forked additional rundll32.exe instances for library.dll and readme.dll. WMI processes wmiadap.exe and wmiprvse.exe ran in parallel, querying hardware profiles as part of the game engine's startup sequence. No injected threads, process hollowing, or anomalous parent-child relationships were identified.

Activity Summary

Process and service actions ⓘ

Processes Created

- "C:\Windows\sysnative\rundll32.exe" "C:\Users\<USER>\Desktop\attachment.dll",#1
- C:\Windows\System32\cmd.exe cmd.exe /C rundll32.exe "C:\Users\user\Desktop\library.dll",#1
- C:\Windows\System32\cmd.exe cmd.exe /C rundll32.exe "C:\Users\user\Desktop\readme.dll",#1
- C:\Windows\System32\conhost.exe C:\Windows\system32\conhost.exe 0xffffffff -ForceV1
- C:\Windows\System32\loaddll64.exe loaddll64.exe "C:\Users\user\Desktop\library.dll"
- C:\Windows\System32\loaddll64.exe loaddll64.exe "C:\Users\user\Desktop\readme.dll"
- C:\Windows\System32\rundll32.exe rundll32.exe "C:\Users\user\Desktop\library.dll",#1
- C:\Windows\System32\rundll32.exe rundll32.exe "C:\Users\user\Desktop\readme.dll",#1
- C:\Windows\System32\rundll32.exe rundll32.exe C:\Users\user\Desktop\library.dll,RUNE
- C:\Windows\System32\rundll32.exe rundll32.exe C:\Users\user\Desktop\readme.dll,RUNE

Shell Commands

- %SANDBOX_DLL_LOADER_AMD64% %SAMPLEPATH% %WORKDIR% 483
- rundll32.exe %SAMPLEPATH%,RUNE
- "PlayGTAV.exe"

Processes Terminated

- %SANDBOX_DLL_LOADER_AMD64% %SAMPLEPATH% %WORKDIR% 483

Figure 6: Processes created and shell commands — %SANDBOX_DLL_LOADER_AMD64% invoking 483 export ordinals and rundll32.exe executing the RUNE named entry point

Activity Summary

- C:\Windows\System32\loaddll64.exe
- C:\Windows\System32\rundll32.exe

Processes Tree

- 2536 - %SANDBOX_DLL_LOADER_AMD64% %SAMPLEPATH% %WORKDIR% 483
 - ↳ 2616 - rundll32.exe %SAMPLEPATH%,RUNE
- 2740 - wmiadap.exe /F /T /R
- 2784 - %windir%\system32\wbem\wmiprvse.exe
- 4312 - "C:\Windows\sysnative\rundll32.exe" "C:\Users\<USER>\Desktop\attachment.dll",#1
- 6512 - C:\Windows\System32\loaddll64.exe loaddll64.exe "C:\Users\user\Desktop\readme.dll"
 - ↳ 1156 - C:\Windows\System32\conhost.exe C:\Windows\system32\conhost.exe 0xffffffff -ForceV1
 - ↳ 6164 - C:\Windows\System32\cmd.exe cmd.exe /C rundll32.exe "C:\Users\user\Desktop\readme.dll",#1
 - ↳ 6740 - C:\Windows\System32\rundll32.exe rundll32.exe "C:\Users\user\Desktop\readme.dll",#1
 - ↳ 3156 - C:\Windows\System32\rundll32.exe rundll32.exe C:\Users\user\Desktop\readme.dll,RUNE

Synchronization mechanisms & Signals ⓘ

Mutexes Created

- \Sessions\1\BaseNamedObjects\Local\SessionImmersiveColorMutex

Figure 7: Process execution tree — loadll64.exe parent chain, conhost.exe console attachment, and WMI hardware enumeration subprocesses

Task 2: Windows Defender Quarantine Auditing & Remediation

Windows Defender generated three distinct threat alerts targeting the GTA V crack directory. A Severe-level classification, Trojan:Win32/Kepavll!rfn, was raised on June 16, 2026 against RUNE64.dll, with quarantine failing — indicating the file could not be isolated, likely because it was held open by another process. Two PUA-level detections followed on June 19, 2026: PUA:Win32/Packunwan targeting RUNE64.dll and PUA:Win32/GameHack targeting PlayGTAV.exe.

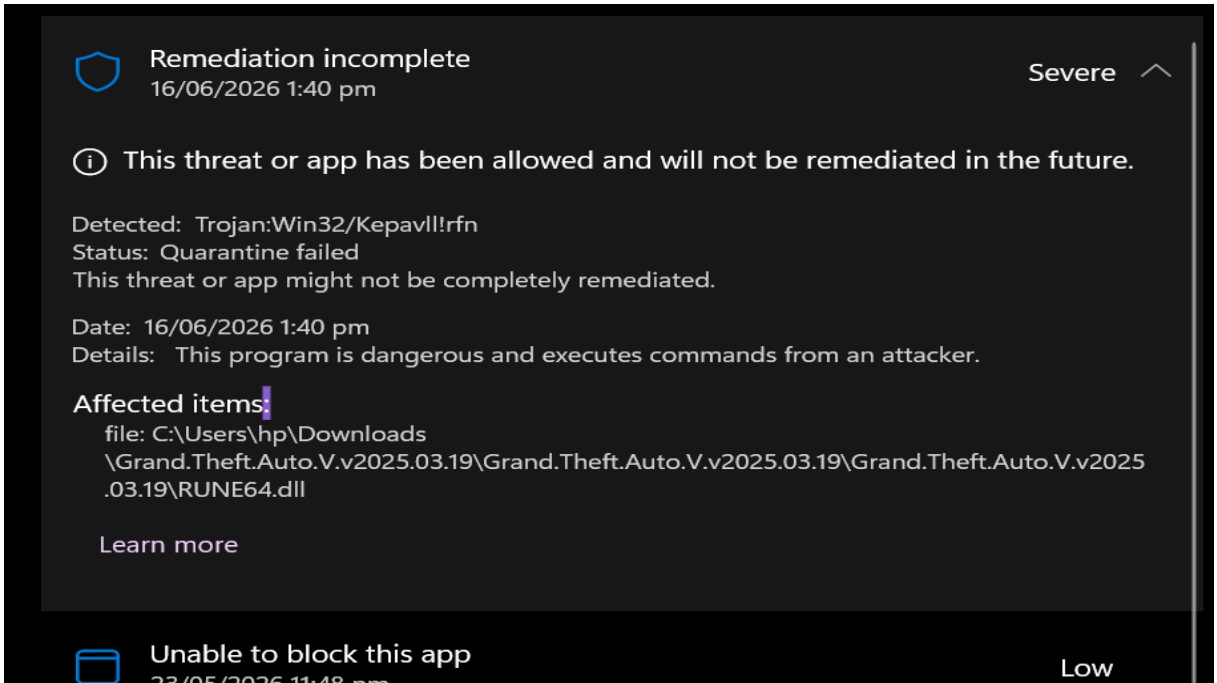


Figure 8: Severe Trojan:Win32/Kepavll!rfn detection with failed quarantine on RUNE64.dll (June 16, 2026)

The Trojan:Win32/Kepavll!rfn classification is a generic heuristic raised when Defender's scan engine encounters binary logic it cannot parse — a direct consequence of VMProtect virtualizing the native x64 assembly. PUA:Win32/Packunwan was triggered by the structurally anomalous PE section layout, while PUA:Win32/GameHack is a standard detection applied to DRM bypass and game crack executables. The table below summarizes the complete threat profile.

File	Threat Name	Severity	Status
RUNE64.dll	Trojan:Win32/KepavII!rfn	Severe	Quarantine failed
RUNE64.dll	PUA:Win32/Packunwan	Low	Quarantined
PlayGTAV.exe	PUA:Win32/GameHack	Low	Quarantined

Because Defender's graphical interface did not expose granular restore controls for the Severe-level Trojan entry, MpCmdRun.exe was invoked from an elevated command prompt to audit the internal quarantine database and restore the flagged binaries to a Defender-excluded directory.

```
cd "C:\Program Files\Windows Defender"
MpCmdRun.exe -Restore -Name "Trojan:Win32/KepavII!rfn"
MpCmdRun.exe -Restore -Name "PUA:Win32/Packunwan"
MpCmdRun.exe -Restore -Name "PUA:Win32/GameHack"
```

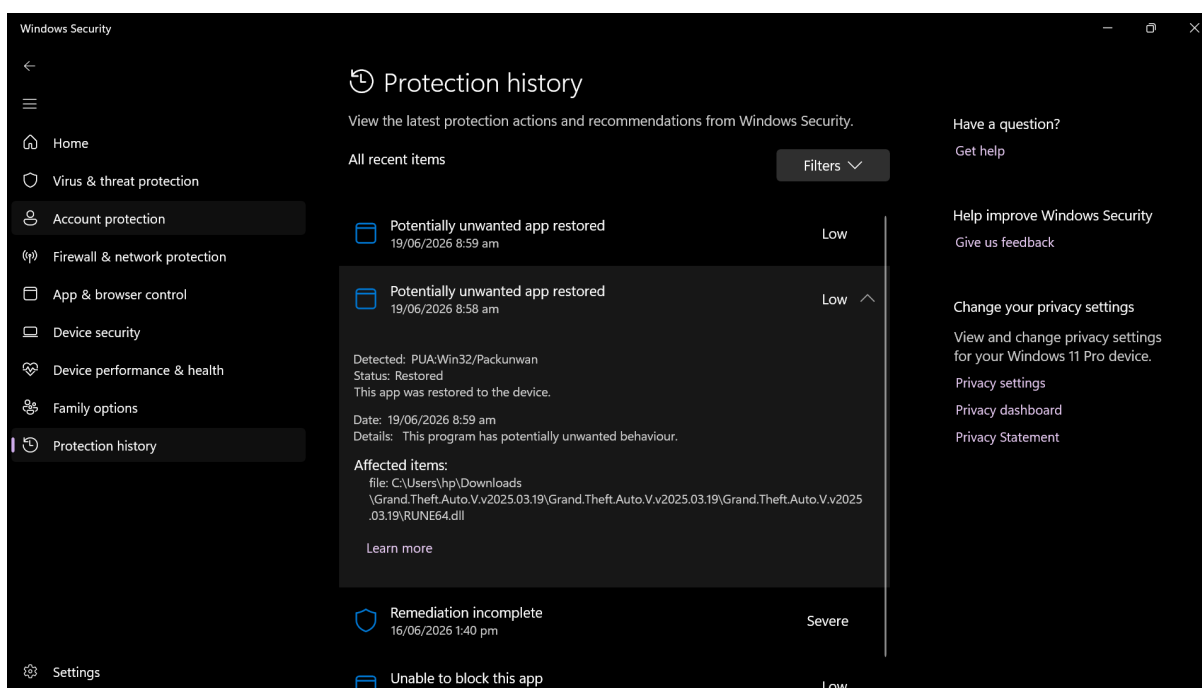


Figure 9: Windows Security Protection History confirming PUA restoration events on June 19, 2026

Task 3: Static PE Analysis with Detect It Easy (DIE)

Following successful restoration, RUNE64.dll was placed in an isolated analysis directory alongside PlayGTAV.exe and loaded into Detect It Easy v3.2.0 for static PE header inspection. The binary presented as a 64-bit AMD64 DLL (PE64) targeting Windows Vista and later, compiled from C/C++ source using Microsoft Visual C++ and linked with Microsoft Linker 14.43. At 11.13 MiB, it is substantially larger than a typical utility DLL — consistent with the size overhead introduced by VMProtect's embedded virtual machine interpreter.

DIE's signature scan identified two high-significance findings that directly account for every antivirus detection generated in this analysis. The first was VMProtect 1.70 at maximum protection level. VMProtect is a commercial software protection suite that converts native x64 machine code into randomized virtual bytecode, executed at runtime by an embedded proprietary virtual machine. This transformation renders the binary's logic completely opaque to heuristic signature scanners. Defender cannot read the virtualized code and defaults to a Trojan classification as a conservative response to the unreadable payload — the direct origin of the Trojan:Win32/Kepavll!rfn detection.

The second finding was an anomalous PE section structure. DIE flagged a generic packer signature resulting from section collision: a custom .rune section, a compressed Section #7 labeled .rune2, and Section #0 (.text) carrying an incorrect offset and size relative to standard MSVC output. These custom sections house the encrypted Steam emulation hooks that intercept and replace Steamworks DRM calls with cracked equivalents. Their non-standard naming and compressed layout conflict with expected PE format conventions — the structural basis for both the PUA:Win32/Packunwan designation and the DIE packer detection.

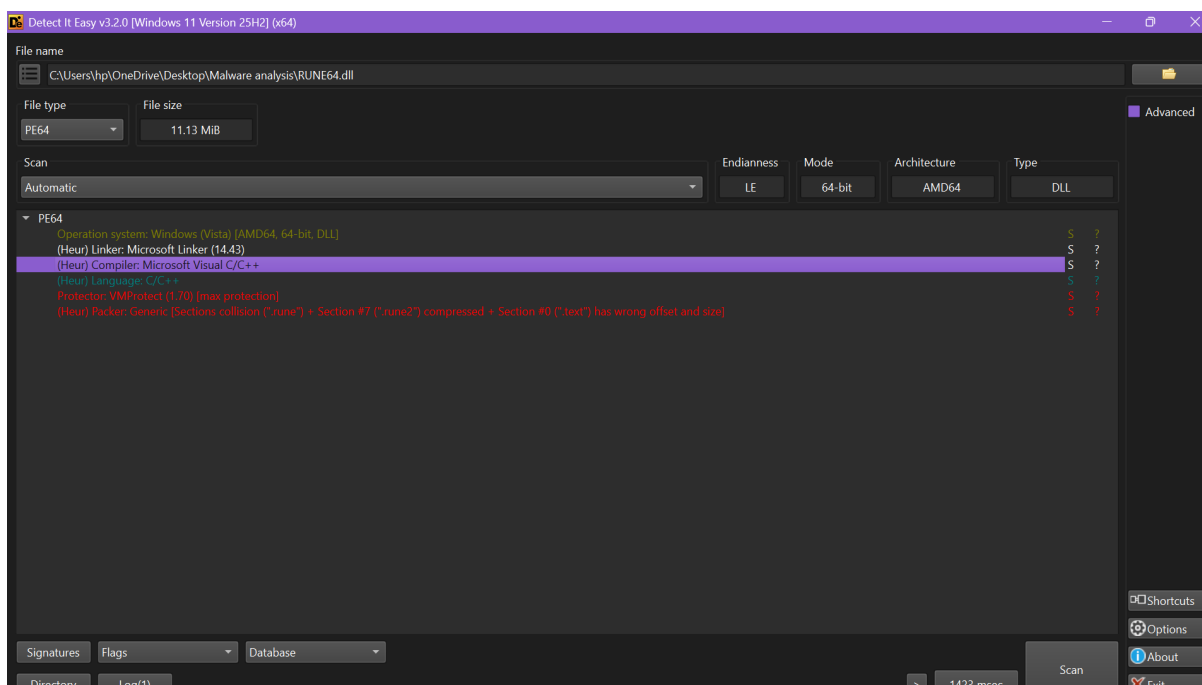


Figure 10: Detect It Easy — VMProtect 1.70 and anomalous .rune/.rune2 PE sections identified in RUNE64.dll

VMProtect is a legitimate commercial tool used by software vendors for copy protection and by scene release groups for crack packaging. Its presence in a DRM bypass DLL, while reliably triggering automated threat detection pipelines, is architecturally expected and does not constitute evidence of malicious authorship.

Summary

All three analysis phases returned consistent findings. VirusTotal's behavioral sandbox environments confirmed the absence of dropped secondary payloads, unauthorized Command and Control communication, and host persistence mechanisms. Every DNS resolution resolved to Microsoft CDN infrastructure. The process execution tree matched the expected invocation pattern for a scene crack precisely, with no anomalous parent-child relationships, injected threads, or process hollowing observed.

Windows Defender's threat classifications were traced to two structural properties identified by Detect It Easy: VMProtect 1.70 obfuscation at maximum protection — which renders the binary opaque to the scan engine and triggers a conservative Trojan heuristic — and custom PE sections (.rune, .rune2) with non-standard layout that conflict with expected MSVC format signatures, producing the Packunwan and GameHack PUA designations. Neither property constitutes malicious behavior; both are standard architectural characteristics of high-tier

DRM emulation tooling. The final analyst verdict is a confirmed false positive: RUNE64.dll and PlayGTAV.exe exhibit no behavioral indicators of host compromise.